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17 December 1971
SUPERSEDING
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30 March 1970

MILITARY SPECIFICATION

BODY ASSEMBLY: 10516210
(PERISCOPE:M32)

1. SCOPE

1.1 This specification covers a modular unit of the Tank Gunner's M32 Periscope, herein referred to as "body assembly".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F8565556	Pressure Tester, Two-Station
F10516210	Body Assembly
F11732109	Gage Interchangeability, Maximum
F11732110	Gage Interchangeability, Minimum

FSC 1240

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PACKAGING DATA SHEET

10516210 Packaging of Body Assembly 10516210

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The body assembly shall be manufactured in accordance with Drawing F10516210 and drawings pertaining thereto.

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Environmental.

3.4.1 Storage temperature.- Prior to meeting the requirements of 3.6 to 3.17 inclusive, the body assembly shall show no evidence of physical failure such as glass fracture or breakage, bonding or casting failure due to storage at temperatures ranging from -80° to +160°F.

3.4.2 Operating temperatures.- The body assembly shall meet the applicable requirements of 3.15 while exposed and thermally stabilized at ambient of -40° and +150°F.

3.4.3 Vibration.- The body assembly shall show no evidence of physical failure due to being vibrated in a vertical plane at a constant frequency of 30 cycles per second (cps) with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. Following the vibration test, the body assembly shall meet all remaining requirements herein.

3.5 Cleanliness.

a. Defects.- There shall be no evidence of moisture, grease,

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fingerprints, condensates, fractures, and adhesive separations on any glass component when viewing through either end.

b. Dirt.- Inspection for dirt and foreign particles (classed as dirt) shall be made from the eye end only. The maximum size of any particle appearing on the reticle shall be a reticle line width. The maximum number of dirt particles appearing within a central field of view of 50 mils diameter shall be three. The maximum number of dirt particles appearing outside this field of view shall be three. The minimum angular separation between dirt particles shall be 15 mils. Dirt particles smaller than 1/4 reticle line width shall be ignored regardless of distribution.

3.6 Sealing.- The body assembly shall show no evidence of leakage in excess of 0.2 pounds per square inch (psi) when subjected to an internal pressure of 5. ± 0.10 psi for a minimum of 2 hours. The interior atmosphere of the accepted body assembly shall contain nitrogen having a dew point at least as low as minus 25°F at atmospheric pressure. The body assembly shall then be sealed with the dry nitrogen specified to an internal pressure of between 0.25 and 0.50 psig.

3.7 Orientation.- The requirements of 3.8 to 3.14 inclusive shall be met with the eyepiece focused on the reticle surface.

NOTE: The following references apply throughout this specification

Reference Line "Y", see Drawing F10516210
Datum Plane "X" (Intersection of planes)

3.8 Parallax.- Parallax between the reticle and the image of a target at 1200 meters shall not exceed 0.1 mil as adjusted by Eccentric 8205635.

3.9 Collimation.- These requirements shall be met prior to meeting the requirements of 3.4.3.

3.9.1 Previbration setting.- An incident line of sight shall be provided which is perpendicular to Datum Plane "X" within 0.20 mil. Coincidence of this line of sight with the instrument reticle shall be established by rotation of the boresight knobs.

3.9.2 Reticle movement.- Reticle (boresight cross) movement shall be measured against an external target from the collimated setting of 3.9.1 to meet the requirements of 3.9.2.1 and 3.9.2.2.

3.9.2.1 Elevation.- Clockwise rotation of the elevation knob shall move the reticle, in a direction towards the knob, a minimum of 8.5 mils from the collimated setting. Counterclockwise rotation of the elevation knob shall move the reticle, in a direction away from the knob, a minimum of 5.5 mils from the collimated setting.

3.9.2.2 Deflection.- Clockwise rotation of the deflection knob shall

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move the reticle, in a direction towards the knob, a minimum of 7.0 mils from the collimated setting. Counterclockwise rotation of the deflection knob shall move the reticle, in a direction away from the knob, a minimum of 7.0 mils from the collimated setting.

3.9.3 Knob rotation.- Continuous movement of reticle shall be evident until limited by positive engagement of clockwise and counterclockwise stops in each boresight knob.

3.9.4 Line of sight variation.- Variation in the line of sight in elevation, as the deflection knob is rotated by at least 7.0 mils in each direction from the position established in 3.9.1, shall not exceed 0.1 mil (total variation not to exceed 0.2 mil).

3.9.4.1 Reticle shift.- Disengagement or reversal of either knob shall not cause the line of sight to shift by more than 0.1 mil in a direction perpendicular to the intended motion.

3.9.5 Boresight backlash.- Backlash in each boresight mechanism shall not exceed 0.1 mil.

3.9.6 Collimation change due to vibration.- This requirement shall be met subsequent to meeting the requirements of 3.4.3. The incident line of sight previously established shall coincide, within 0.1 mil (including fixture error), with the instrument reticle.

3.10 Image tilt.- The image of a target line which is originally perpendicular to the Reference Line "Y" shall emerge perpendicular to Reference Line "Y" within 1.0 degree of arc.

3.11 Parallelism of reticle and image.- The image of a target line which is originally perpendicular to Reference Line "Y" shall emerge parallel, within 30 minutes of arc to the reticle line.

3.12 Objective focusing range.- Rotation of Eccentric 8205635 shall cause the objective assembly to travel a minimum of 0.020 inch in each direction from the position established in 3.8.

3.13 Dioptral adjustment.

3.13.1 Zero setting.- This requirement shall be met while observing the projected image through a dioptometer set for zero diopter. The diopter scale shall indicate zero within 1/4 diopter when the reticle is brought into focus.

3.13.2 Excursion range.- The diopter movement shall be adjustable through a minimum range of -3.0 to +3.0 diopters as read on the diopter scale.

3.13.3 Diopter backlash.- Backlash in the diopter movement shall not exceed 1/8 diopter as read on the diopter scale.

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3.14 Resolution.- The resolution shall be 6.0 seconds of arc or less when viewing with an auxiliary telescope having magnification of at least 3.0 power. Resolution consists of detectable line structure and proper line count in all four meridians. Eyepiece focus shift between any two meridians shall not exceed 0.25 diopter.

3.15 Operability.

3.15.1 Disengagement of boresight knob locks.- The direct force required to disengage each boresight knob from the lock position shall be within the limits of 2.0 to 5.0 pounds (lbs) when checked at standard temperature. When checked at the temperature condition specified in 3.4.2, the direct force shall not exceed 9.0 lbs.

3.15.2 Rotation of boresight knobs.- The running torque required to rotate each boresight knob through the range specified in 3.9.3 shall be within the limits of 0.5 and 5.0 pound inches (lb-ins) when checked at standard temperature. When checked at the temperature condition specified in 3.4.2, the running torque shall not exceed 8.0 ins.

3.15.3 Diopter movement.- The running torque required to rotate the diopter movement through the range specified in 3.13.2 shall be within the limits of 1.5 and 6.0 lb-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 15 lb-ins.

3.15.4 Focusing lock.- This requirement shall be met with a 40 ounce inches (oz-in) torque applied to the locking screw on Housing 8634490. There shall be no rotation of Eccentric 8205635 when 48 oz-in torque is applied to the eccentric.

3.15.5 Objective focusing mechanism.- This requirement shall be met with the locking screw backed off a minimum of two full revolutions from the locked position established in 3.15.4. The running torque required to rotate Eccentric 8205635 through a minimum of one full revolution in each direction shall not exceed 30 oz-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 50 oz-ins.

3.16 Reticle illumination.- With a 24 volt direct current source energizing the reticle light, the marking shall be visible when observed in a dark room.

3.17 Interchangeability.- The body assembly shall be capable of insertion and removal from the gages specified in 4.6.7.16.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as

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otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three body assemblies plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests

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related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. In addition to 100% inspection as specified in 4.3.2.2, the tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejected by item or lot.

4.3.2.2 100 Percent inspection.- The following requirements and related tests shall be subjected to 100% inspection:

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
1.	Line of sight variation	3.9.4	4.6.7.5
2.	Reticle shift	3.9.4.1	4.6.7.6
3.	Dioptral adjustment zero setting	3.13.1	4.6.7.11.1
4.	Resolution	3.14	4.6.7.13
5.	Reticle illumination	3.16	4.6.7.15
6.	Interchangeability	3.17	4.6.7.16

TABLE I - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65% Defective		
101.	Pre-vibration collimation setting	3.9.1	4.6.3
102.	Vibration	3.4.3	4.6.4
103.	Collimation change due to vibration	3.9.6	4.6.7.1
104.	Cleanliness	3.5	4.6.5
105.	Sealing	3.6	4.6.6
106.	Parallax	3.8	4.6.7.2
107.	Reticle movement, elevation	3.9.2.1	4.6.7.3
108.	Reticle movement, deflection	3.9.2.2	4.6.7.3.1
109.	Knob rotation	3.9.3	4.6.7.4
110.	Boresight backlash	3.9.5	4.6.7.7
111.	Image tilt	3.10	4.6.7.8
112.	Parallelism of reticle and image	3.11	4.6.7.9
113.	Objective focusing range	3.12	4.6.7.10
114.	Dioptral adjustment excursion range	3.13.2	4.6.7.11.2
115.	Diopter backlash	3.13.3	4.6.7.12

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TABLE I - CLASSIFICATION OF DEFECTS Cont'd

		REQUIREMENT	TEST PROCEDURE
116.	Operability, disengagement of boresight knob locks	3.15.1	4.6.7.14.1
117.	Operability, rotation of boresight knobs	3.15.2	4.6.7.14.2
118.	Operability, diopter movement	3.15.3	4.6.7.14.3
119.	Operability, focusing lock	3.15.4	4.6.7.14.4
120.	Operability, objective focusing mechanism	3.15.5	4.6.7.14.5

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One body assembly shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.2	Applicable Drawing - Visual
302.	General specification	3.3	MIL-F-13926 - MIL-O-13830 - Visual

NOTE: The tests in Table II shall be conducted at a temperature between 60° and 90°F.

4.4.2 Environmental.- Three body assemblies shall be selected at random as samples from each 50 produced, or from each months production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperatures (-80° to +160°F)	3.4.1	4.6.1

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TABLE III (Cont'd)

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Operability, disengagement of boresight knob locks	3.4.2, 3.15.1	4.6.2, 4.6.7.14.1
305.	Operability, rotation of boresight knobs	3.4.2, 3.15.2	4.6.2, 4.6.7.14.2
306.	Operability, diopter movement	3.4.2, 3.15.3	4.6.2, 4.6.7.14.3
307.	Operability, focusing lock	3.4.2, 3.15.4	4.6.2, 4.6.7.14.4
308.	Operability, objective focusing mechanism	3.4.2, 3.15.5	4.6.2, 4.6.7.14.5

NOTE: The tests for Characteristics 304 through 308 inclusive, shall be conducted at -40° and at +150°F.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- Special testing equipment to test sealing of the body assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.2 Performance.- Special testing equipment to test performance of the body assembly shall conform to the following requirements:

a. The testing equipment shall be capable of securely mounting the body with Datum Plane "X" in a vertical plane and Reference Line "Y" in a horizontal plane within 10 seconds. The necessary mounting features (keyways and locating surfaces) shall be incorporated into the testing equipment.

b. A target projector collimator having a clear aperture of at least 2.0 inches shall be a part of the test equipment. The target projector collimator shall be positioned and mounted so that the optical axis of the collimator is in alinement with the line of sight emitting from the central aperture (geometric center) of the body assembly objective. The reticle of the target projector collimator shall have an appropriate calibrated crossline target test pattern capable of measuring the optical accuracies of the body assembly. The vertical reticle line of the target test pattern shall be adjusted plumb. The optical axis of the target projector collimator shall be alined parallel with respect to Reference Line "Y" and autocollimated within 10 seconds to Datum Plane "X" of the body assembly.

c. A calibrated observation telescope of at least three power (3X) shall be a part of the test equipment. The observation telescope shall be positioned and mounted so that the optical axis of the telescope is in alinement with the line of sight emitting from the central aperture (geometric center) of the body assembly eyepiece. The reticle of the telescope shall have a crossline reticle pattern with tolerance lines to determine image tilt. The optical axis of the telescope shall be alined to the infrared target projector collimator and autocollimated to Datum Plane "X" within 10 seconds.

d. Autocollimation and alinement shall be accomplished with the body assembly removed from the line of sight through the target projector collimator and the observation telescope. A set-up device containing a penta prism having a deviation of ± 5 seconds (calibrated) shall be utilized for alinement and autocollimation of the target projector collimator and the observation telescope. The necessary mounting features (keys and locating surfaces) shall be incorporated in the set-up device.

e. All set-up and calibrated equipment with complete provisions necessary for certification and proof of accuracy of the test equipment shall be provided and become a part of the testing equipment.

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f. Upon completion of the fixture set-up and alinement procedures, the observation telescope shall be capable of being removed from the optical line of sight of the test equipment when not required in the performance tests.

4.6 Test methods and procedures.

4.6.1 Storage temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see Note below) to -80°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually raised (see Note below) to +160°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F), and the body assembly removed from the test chamber. The body assembly shall then be subjected to the visual, tactile and audible examination of 4.6.4. The body assembly shall conform to the requirements of 3.4.1.

NOTE: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see Note in 4.6.1) to -40°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, and while at 40°F, the body assembly shall be subjected to the tests in 4.6.7.14.1 to 4.6.7.14.5, inclusive to determine compliance with the applicable requirements of 3.15.1 to 3.15.5 inclusive, for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (see Note in 4.6.1) to +150°F. The body assembly shall remain at this temperature for a period of 5 hours. At the completion of this 5 hour period and while at +150°F, the body assembly shall again be subjected to the test in 4.6.7.14.1 to 4.6.7.14.5 inclusive, to determine compliance with the applicable requirements of 3.15.1 to 3.15.5 inclusive, for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see Note in 4.6.1) to room ambient temperature (+60° to +90°F) and the body assembly removed from the test chamber. Body assemblies which have passed the above tests shall be subjected to the test in 4.6.3.

4.6.3 Collimation, previbration setting.- Prior to the vibration test of 4.6.4, the body assembly shall be positioned as in 4.6.7 to insure conformance to 3.9.1. While viewing through the body assembly eyepiece,

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obtain coincidence of the body assembly reticle boresight cross and the projected target image crosslines, by means of the boresight knobs. When coincidence has been obtained, slip the scales of the boresight knobs to a setting of 4 and recheck to insure coincidence is not disturbed. Record these settings for use in the test in 4.6.7.1. Without disturbing the boresight knob scale settings of (4), the body assembly shall then be subjected to the test in 4.6.4.

NOTE: Insure that knob scales are not disturbed (re-slipped) from their original setting of (4) throughout remainder of tests.

4.6.4 Vibration.- This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope this vibration test will not be required when the complete periscope is to be vibrated. Standard vibrating equipment capable of proving the amplitude and frequency specified in 3.4.3 shall be utilized for this test. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The body assembly shall be clamped on the vibrating equipment in its normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing F10516210) by means of an appropriate adapter. The vibration motion shall be in the vertical plane for the time duration specified in 3.4.3. Immediately following the vibration cycle, and prior to the examination for physical defects resulting from poor workmanship and materials, the body assembly shall be subjected to the test in 4.6.7.1. Do not disturb the boresight knob scale settings of (4), during transfer of the body assembly. Subsequent to the test in 4.6.7.1, the body assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.3, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. With the exception of the slip scales, a tactile examination shall be made for functioning of moving parts for undue irregularities of movement, friction or looseness. An audible examination shall also be made of the body assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a body assembly defective. Body assemblies which have passed the above examinations shall be subjected to the remaining tests specified in 4.6.5 through 4.6.7.16 inclusive.

4.6.5 Cleanliness.- The optical surfaces of the body assembly shall be visually examined with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through the eyepiece, inspection for dirt shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 apparent foot candles. The body assembly shall conform to the requirements of 3.5.

4.6.6 Sealing.- This portion of the test shall be performed utilizing a pressure gage capable of reading between 0.25 to 0.50 pound per square inch gage (psig) and a dew point tester. The following test sequence shall

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be adhered to. Check the internal pressure of the body assembly utilizing the pressure gage specified herein. Then, test the dew point of the interior atmosphere of the body assembly by means of the dew point tester. Interior atmosphere of the body assembly shall conform to the requirements of 3.6. Upon satisfactory completion of this portion of the test internal pressurizing of the body assembly shall be performed with special testing equipment conforming to 4.5.2.3.1. Pressurize the body assembly with dry nitrogen gas, having a dew point at least as low as -25°F to 5 pounds per square inch (psi) in accordance with the procedure for internal testing on Drawing F8565556. Retain the applied pressure in the body assembly for the time duration specified in 3.6. Within the first 60 minutes of this time period, the boresight knobs and the diopter adjustment shall be rotated to their clockwise and counterclockwise stops, and with the locking screw on Housing 8634490 backed off from its locked position, the Eccentric 8205635 shall be rotated a minimum of two full revolutions in each direction. At the end of this time period, any evidence of leakage as indicated on the pressure gage of the testing equipment shall not exceed the tolerance specified in 3.6. Upon satisfactory completion of the sealing test, the body assembly shall then be flushed with the dry nitrogen gas specified above. The interior atmosphere of the body assembly shall be periodically sampled through the dew point tester until the atmosphere shows a dew point at least as low as -25°F. When this atmosphere is attained within the body assembly, the body assembly shall then be sealed at the pressure specified in 3.6. Upon completion, readjust the eccentric to the position established in 4.6.7.2 and torque the locking screw as specified in 3.15.4.

4.6.7 Orientation.- The following tests shall be performed with the body assembly mounted on a special testing equipment conforming to 4.5.2.3.2. The target projector collimator shall be adjusted for 1200 meters and unless otherwise specified, the eyepiece diopter knob shall be set for best focus of the body assembly reticle and the target image.

4.6.7.1 Collimation change due to vibration.- Observe through the body assembly eyepiece and adjust by means of the boresight knobs to obtain coincidence of the body assembly reticle boresight cross with the target image. Boresight knob adjustments shall not exceed the tolerance specified in 3.9.6, as measured on the boresight knob scales.

NOTE: Do not slip knob scales.

4.6.7.2 Parallax.- Perform this test with the body assembly diopter adjustment knob adjusted for best focus of the body assembly reticle. Parallax requirements shall be met when the eccentric is adjusted as specified in 3.8. While viewing through the body assembly eyepiece parallax will be recognized by any apparent displacement of the intersecting point of the body assembly reticle boresight mark with respect to the intersecting point of the vertical and horizontal target line image when the observer's head is moved from side to side or up and down. Displacement between the body assembly reticle and target image shall not exceed the tolerance specified in 3.8. Upon satisfactory completion of this test, perform the tests specified in 4.6.7.2.1 and 4.6.7.10.

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4.6.7.2.1 Parallax adjustment.- If adjustment is necessary, a standard type screw driver that properly fits the screw slot of the locking screw on Housing 8634490 and Eccentric 8205635 shall be utilized for performing this adjustment. Back off the locking screw a minimum of two revolutions from its locked position. Then while performing the test in 4.6.7.2, rotate the eccentric clockwise or counterclockwise until the parallax requirement specified therein has been met. When the adjustment is complete, apply the applicable torque value specified in 3.15.4 to the locking screw utilizing the torque wrench specified in 4.6.7.14.4.

4.6.7.3 Reticle movement, elevation.- Establish coincidence of the body assembly reticle boresight cross and the projected target image crosslines by means of the boresight knobs (do not slip knob scales). While observing the boresight cross (intersecting point), rotate the elevation knob in the clockwise direction. The boresight cross shall move as specified in 3.9.2.1 as observed and measured on the target. Re-establish coincidence of the reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross, rotate the elevation knob in the counterclockwise direction. The boresight cross shall move as specified in 3.9.2.1 as observed and measured on target.

4.6.7.3.1 Reticle movement, deflection.- Perform this test utilizing the procedure in 4.6.7.3 except, the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight cross (intersecting point) movements shall meet the requirements specified in 3.9.2.2 as observed and measured on the target.

4.6.7.4 Knob rotation.- Perform this test in conjunction with the tests in 4.6.7.3 and 4.6.7.3.1. During rotation of each boresight knob (elevation and deflection), the boresight cross (intersecting point) shall be observed and show no evidence of irregularities (jump, creep or hang up) in direction of travel until limited by engagement of each knob stop (clockwise and counterclockwise) to be in compliance with 3.9.3. While the boresight knobs are at each stop position, back off the knobs for at least 1/4 rotation and then again return the knobs to the stop. Perform this a minimum of 3 times. During these rotations, the knobs shall show no evidence of irregularity of movement (sponginess or friction) while approaching or reaching stop. Contact with the stop shall be solid, such that a distinct click can be heard to verify that reticle travel is being limited only by means of the knob stops and not by means of contact with adjacent components to determine overall compliance with 3.9.3.

4.6.7.5 Line of sight variation.- Perform this test in a manner similar to the test in 4.6.7.3.1. During rotation of the deflection knob, any variation in elevation in the line of sight, through the boresight cross (intersecting point), shall not exceed the values specified in 3.9.4.

4.6.7.6 Reticle shift.- With backlash removed for direction of travel, position the intersecting point of the body assembly reticle boresight mark as in 4.6.7.3. While observing the intersecting point of the

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body assembly reticle boresight mark, disengage the elevation boresight knob from the lock position and proceed to rotate the knob (insure backlash is not introduced) for at least 1/4 turn, then reverse rotation of the knob (backlash included) for at least 1/4 turn. During the disengagement or reversal of the boresight knob, the line of sight through the body assembly reticle boresight mark shall not shift (jump) in a direction perpendicular to the intended motion more than the tolerance specified in 3.9.4.1. In addition to checking shift at this setting, repeat the above at the midpoint and maximum point of travel in each direction for a total of 5 checks overall. These 5 checks shall also be conducted with the deflection knob in compliance with 3.9.4.1.

4.6.7.7 Boresight backlash.- While viewing through the body assembly eyepiece, disengage and rotate the boresight knobs (insure backlash is removed for direction of travel) until a line of sight, through the intersection of the body assembly reticle boresight mark is positioned coincident with the intersecting point of the target image crosslines. With the boresight knobs in the lock position, slip the scales to Index 4 and record the position of the line of sight. Disengage the boresight knobs and continue rotation approximately 1/2 of a revolution then return and engage the knobs to their original lock position setting of Index 4. View through the eyepiece and record the position of the line of sight through the reticle boresight mark. Any difference between the recorded positions of the line of sight for each boresight knob is the amount of backlash. Backlash in each boresight mechanism shall not exceed the tolerance specified in 3.9.5, as measured on the target.

NOTE: Should overtravel occur during rotation of the boresight knobs the test is nullified and shall be repeated.

4.6.7.8 Image tilt.- This test shall be performed with the aid of the observation telescope positioned at the body assembly eyepiece. View through the observation telescope and establish one extremity of the telescope horizontal reticle line coincident to the horizontal target image line. The image of the horizontal target line shall be parallel to the telescope vertical line within the tolerance specified in 3.10, as measured on the observation telescope.

4.6.7.9 Parallelism of image and reticle.- Perform this test while observing through the body assembly eyepiece with the unaided eye. Position one extremity of the body assembly vertical reticle line coincident to the vertical target image line. The body assembly vertical reticle line shall be parallel to the vertical target image line within the tolerance specified in 3.11, as measured on the target.

4.6.7.10 Objective focusing range.- This test shall be performed subsequent to the test in 4.6.7.2. A dial indicator, graduated in .001 of an inch, mounted on an appropriate stand, a supporting medium capable of receiving the dial indicator as well as the body assembly to be tested and the screw driver specified in 4.6.7.2.1 shall be utilized for this test. With the body assembly firmly secured on the supporting medium,

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position the dial indicator so that its contact point touches the face of the movable objective tube of the body assembly and set the dial to read zero with the indicating hand. Back off the locking screw of Housing 8634490 a minimum of two full revolutions from its locked position. Starting from the zero reading of the dial indicator, rotate the eccentric until the objective assembly has traveled the minimum distance specified in 3.12, as measured on the dial indicator. Return the indicating hand of the dial indicator to zero by means of the eccentric and repeat this travel in the opposite direction. There shall be sufficient travel in each direction to meet the minimum requirements specified in 3.12, as measured on the dial indicator. Upon completion of this test, repeat the parallax adjustment procedure specified in 4.6.7.2.1. The body assembly shall then be subjected to the test in 4.6.7.14.4.

4.6.7.11 Dioptral adjustment.

4.6.7.11.1 Zero setting.- A calibrated dioptometer with a magnification of at least 3 power (3X) shall be utilized for measuring the accuracy of the zero diopter setting of the diopter scale. Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Position the dioptometer at the eyepiece of the body assembly. While viewing through the dioptometer, obtain best focus of the body assembly reticle by adjusting the body assembly diopter knob. The diopter scale shall indicate zero within the tolerance specified in 3.13.1.

4.6.7.11.2 Excursion range.- Perform this test in conjunction with the test in 4.5.6.11.1. From the best focus of the body assembly reticle established in 4.6.7.11.1, rotate the diopter adjustment knob to its clockwise stop, note the diopter setting on the scale, then rotate the diopter knob to its counterclockwise stop, note diopter setting on the scale. The diopter movement shall meet the requirements specified in 3.13.2, as measured on the diopter of the body assembly.

4.6.7.12 Diopter backlash.- Perform this test utilizing the dioptometer specified in 4.6.7.11.1. The dioptometer shall be zeroed and adjusted as specified therein. Starting from the stop position rotate the body assembly diopter adjustment knob clockwise to the "0" position without reversing direction of rotation. Position the dioptometer at the eyepiece and obtain best focus of the body assembly reticle by adjustment of the dioptometer diopter scale. Record the dioptometer scale reading. Rotate the body assembly diopter adjustment knob clockwise to its stop position, then return it counterclockwise to the "0" position without overtravel. Again obtain best focus of the body assembly reticle by adjustment of the dioptometer diopter scale. The difference between the first and second dioptometer diopter scale readings shall not exceed the value specified in 3.13.3.

NOTE: Should overtravel occur during rotation of the diopter adjustment knob to the "0" position, the test is nullified and shall be repeated.

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4.6.7.13 Resolution.- The test equipment specified in 4.6.7 is not required for this test. Perform this test utilizing the dioptometer specified in 4.6.7.11.1. The dioptometer shall be zeroed and adjusted as specified therein. A resolving power chart on a wall target representing the angular subtense for six seconds of arc and a supporting medium capable of receiving the dioptometer (as well as the body assembly to be tested) shall be a part of the test equipment. Construction of the resolving power chart shall be as specified in the "Resolution Test" of MIL-O-13830 and shall contain four line sets as shown in Figure I therein. Place the body assembly on the supporting medium and position the dioptometer at the eyepiece. While viewing through the dioptometer, obtain best focus of the numeral in the center of the resolving power chart by adjusting the body assembly diopter knob, then obtain best focus of each of the four meridians by adjustment of the dioptometer objective scale. Any change in the diopter readings by adjustment of the dioptometer objective scale to obtain the detectable line structure and proper line count in all four meridians shall be recorded, as measured on the dioptometer objective scale. All four meridians shall have the detectable line structure and proper line count to conform with3.14 and the difference between the dioptometer scale readings for any two meridians shall not exceed the value specified therein.

4.6.7.14 Operability.

4.6.7.14.1 Disengagement of boresight knob locks.- Standard measuring equipment capable of measuring the force values specified in 3.15.1 shall be utilized from this test. The force required to disengage each boresight knob from the lock position shall conform to the applicable requirements of 3.15.1 for the temperatures indicated.

4.6.7.14.2 Rotation of boresight knobs.- Standard torque measuring equipment capable of measuring the torque values specified in 3.15.2 shall be utilized for this test. An appropriate adapter, to accommodate the boresight knobs shall be used with the torque measuring equipment. Torque shall be measured while rotating each boresight knob through the range specified in 3.15.2. The running torque required to rotate each boresight knob shall conform to the applicable requirements of 3.15.2 for the temperatures indicated.

4.6.7.14.3 Diopter movement.- Standard torque measuring equipment capable of measuring the torque values specified in 3.15.3 shall be utilized for this test. An appropriate adapter, to accommodate the diopter knob shall be used with the torque measuring equipment. Torque shall be measured while rotating the diopter movement through the range specified in 3.15.3. The running torque required to rotate the diopter knob shall conform to the applicable requirements of 3.15.3 for the temperatures indicated.

4.6.7.14.4 Focusing lock.- This test shall be performed in sequence to the test in 4.6.7.10. A dial indicator, graduated in increments of .0001 of an inch, mounted on an appropriate stand, the supporting medium specified in 4.6.7.10 and a torque wrench capable of applying the specified

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torque values in 3.15.4, together with a standard type driver bit that properly fits the screw slot of the locking screw and eccentric, shall be utilized for performing this test. Secure the body assembly firmly on the supporting medium. With the specified torque applied to the locking screw, position the dial indicator so that its contact point touches the face of the movable objective tube of the daylight body and set the dial to read zero. Apply, in a clockwise direction, the specified torque to the eccentric, then observe the dial indicator reading. Then apply, in a counterclockwise direction, the specified torque to the eccentric and again observe the dial indicator reading. To conform to the requirements of 3.15.4, there shall be no change in the established dial indicator zero reading when observed after each torque application.

4.6.7.14.5 Objective focusing mechanism.- Perform this test utilizing the test equipment specified in 4.6.7.14.4, except the dial indicator is not required. The locking screw shall be backed off a minimum of two full revolutions from its locked position. Torque shall be measured while rotating the eccentric through the range specified in 3.15.5. The running torque required to rotate the eccentric shall conform to the applicable requirements of 3.15.5 for the temperatures indicated. Upon completion of this test, return the eccentric to the position in 3.8 and torque the locking screw as specified in 3.15.4.

4.6.7.15 Reticle illumination.- This test shall be performed in a dark area. The voltage required to illuminate the reticle lamp shall be supplied from a power source through a control device that shall be capable of supplying and maintaining the required voltage in 3.16. With the reticle lamp energized, the illuminated reticle marking shall be clearly defined when observed through the body assembly eyepiece to determine conformance with 3.16. Should the brightness of a reticle defect be greater than the reticle lines, the defect shall be cause for rejection.

4.6.7.16 Interchangeability.- This test shall be performed utilizing the maximum and minimum interchangeability gages in Drawing F11732109 and F11732110 respectively, to determine compliance with 3.17.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10516210.

6. NOTES

6.1 Intended use.- The Body Assembly:10516210, is a daylight modular unit of the Tank Gunner's Periscope:M32 primarily used in the fire control system of the 105MM Gun Tank:M60 Series.

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6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable packaging data sheet (see 5.1).
d. Applicable stock number.
e. Provisions for first article testing.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A321

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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